INF201 - Lecture 7

19th Oct 2026

Topics

- Data containers
- Records
- NumPy arrays
- Pandas DataFrames

Representing data in Python

- How do we organize data that we want to work with in our code?
  - Keep control: is that value the temperature at noon or the wind speed at midnight?
  - Select data according to different criteria: by name, by location, ...
  - Exchange data between different parts of a large program
  - Perform operations efficiently on large data
- Keeping control is essential
  - If we lose control of what is what, our code will produce incorrect results
  - **Wrong results are worse than no results**: if the program crashes, we at least know something is wrong!
- Code lives long and passes through many hands
  - Code should be self-documenting
  - Data should be self-documenting
  - Example: Weather data from frost.met.no

```
[{'sourceId': 'SN17850:0',
'referenceTime': '2021-01-01T00:00:00.000Z',
'observations': [{'elementId': 'mean(air_temperature P1D)',
                  'value': -0.3,
                  'unit': 'degC',
                  'level': {'levelType': 'height_above_ground',
'unit': 'm', 'value': 2},
                  'timeOffset': 'PT0H',
                  'timeResolution': 'P1D',
                  'timeSeriesId': 0,
                  'performanceCategory': 'C',
                  'exposureCategory': '1',
                  'qualityCode': 0},
```

- We will first look at:
  - Standard containers for data: lists, dictionaries, sets
  - NumPy arrays
  - Pandas dataframes
- The next weeks:

- Object-oriented programming as a technique for consistent representation of and operations on data

Standard containers: Lists, dictionaries, sets

- We often need to process collections of data
  - students in a class
  - series of weather measurements
  - payments in a shop
- Frequently, several items of information are related
  - name, courses taken, grades achieved of a student
  - temperature, humidity, precipitation, wind speed at a given hour
  - items, quantities and prices of goods purchased
- The number of data items we have will change over time
  - we cannot work with fixed variables, e.g.,
    ```
    inf201_student1_name = 'Nikolai'
    inf201_student2_name = 'Amalie'
    ...
    inf201_student159_name = 'Henrik'
    ```
  - we need flexible *containers*
- General properties of containers
  - We can insert data
  - We can remove data
  - We can access data
  - We can query if an item is in a container
- These general properties apply not only to containers in Python, but also in other programming languages
- Different containers have different specific properties

Feature	List (list)	Dictionary (dict)	Set (set)
Structure	Ordered sequence	Key-value pairs	Unordered collection of unique items
Duplicates	Allowed	Keys: unique / Values: allowed	No duplicates allowed
Lookup Method	By integer index (`lst[0]`)	By key (`dct["key"]`)	By membership check (`x in st`)
Lookup Time	$O(1)$ by index, $O(n)$ by value	$O(1)$ average by key	$O(1)$ average by value
Syntax	`[1, 2, 3]`	`{"a": 1, "b": 2}`	`{1, 2, 3}`

When to use which?

Use a list when:

- Order matters: You need to preserve the specific sequence of items (e.g., a queue, steps in a workflow, or chronological logs).
- Duplicates are valid: The same value can appear multiple times (e.g., daily temperature readings `[72, 75, 72]`).
- You access elements by position: You know items will be fetched via integer indices (`data[0]`) or sliced (`data[1:4]`).

Use a dictionary when:

- Data consists of key-value associations: You are modeling structured records, config settings, JSON-like objects, or database rows (e.g., `{"name": "Alice", "id": 101}`).
- You need fast lookups by explicit label: You want to retrieve data using a unique key or identifier instead of an arbitrary numerical position.
- Frequency counting: Tracking occurrences of items (e.g., mapping words to their count).

Use a set when:

- Uniqueness is mandatory: You need to automatically drop or prevent duplicate entries (e.g., unique user IDs).
- Frequent membership testing: You regularly run `if item in collection` on large datasets. Checking `item in set` is instantaneous ($O(1)$), whereas item in list requires checking elements one by one ($O(n)$).
- Mathematical operations: You need set logic like finding common elements (intersection `&`), combining items (union `|`), or finding differences (`-`).

Records and fields

In [ ]:
```
students = [{'given_name': 'Per', 'family_name': 'Olsen', 'programme': 'M-DV',
             'courses': ['INF201', 'INF221', 'INF250']},
            {'given_name': 'Kari', 'family_name': 'Hansen', 'programme': 'M-TDV'
             'courses': ['DAT300', 'DAT390', 'FYS388']}]
students
```

- We have come aross the term records before , for example in when writing data to json files
```
result = overdue.to_dict(orient="records")
with open(config["output_file"], "w") as f:
    json.dump(result, f, indent=2)
```
- Our student data contains one record per student.
  - A record comprises information about an entity, e.g., one student.
  - If we had student data in a spreasheet, a record would usually be a row.
  - Standard databases also store records in rows.
  - If we used cardboard cards, we would have one card per student as record.
- Each record has one or more fields
  - A field contains a single piece of information, e.g., family name, phone number, etc.

    - In a spreadsheet or database, fields correspond to columns.
    - On a cardboard card, for each field we would have a box to fill in the data.
- When representing students as a list of dictionaries
    - each dictionary is the record for one student, and
    - dictionary entries map field names to field entries.

Design question: List of dicts or dicts of dicts?

Alternative A: List of records

In [ ]:
```
students = [{'given_name': 'Per', 'family_name': 'Olsen', 'programme': 'M-DV',
             'courses': ['INF201', 'INF221', 'INF250']},
            {'given_name': 'Kari', 'family_name': 'Hansen', 'programme': 'M-TDV
             'courses': ['DAT300', 'DAT390', 'FYS388']}]
```

In [ ]:
```
# How would you extract all information on the person with family name "Olsen"?
```

Alternative B: Mapping family names to data

In [ ]:
```
students_by_fam = {'Olsen': {'given_name': 'Per', 'programme': 'M-DV',
                             'courses': ['INF201', 'INF221', 'INF250']},
                   'Hansen': {'given_name': 'Kari', 'programme': 'M-TDV',
                              'courses': ['DAT300', 'DAT390', 'FYS388']}}
```

In [ ]:
```
# How would you extract the list of courses that Olsen attends?
```

Alternative C: Mapping family and given name to data

In [ ]:
```
students_by_name = {('Olsen', 'Per'): {'programme': 'M-DV',
                                       'courses': ['INF201', 'INF221', 'INF250']
                    ('Hansen', 'Kari'): {'programme': 'M-TDV',
                                         'courses': ['DAT300', 'DAT390', 'FYS388
```

In [ ]:
```
# How would you extract the information about Kari Hansen?
```

Which one do I use?

- Are there principal (will not work!) problems with some alternatives?
- Do some alternatives contain more or less information than others?
- Do they provide more or less flexibility?
- Do they provide advantages?

NumPy

- Numerical Python
- The foundational library for scientific and numerical computing in Python
- Many other Python packages are built on NumPy:
    - Pandas - data analysis
    - SciPy - scientific computing

    - Matplotlib - plotting
    - Biopython - bioinformatics
- The data containers used by NumPy are called NumPy arrays
- The Visual Guide to NumPy by Lex Maxomiv is rally nice illustrated guide to NumPy
- The Scientific Python lectures covers NumPy and other scientific computing topics
- Chapter 4 of *Scientific Computing for Chemists with Python* by Charles J. Weiss

NumPy arrays

- Python lists can contain "anything"
    - Flexible, but requires much space and time
- NumPy
    - Arrays containing a fixed number of elements of the same data type
    - Compact and efficient storage
    - Many mathematical operations can be applied to entire array
- Numpy arrays can be 1D (vectors), 2D (matrices), or multi-dimensional tensors
- In general: NumPy arrays have datatype `ndarray` (n-dimensional array)

Feature	Python list	NumPy array
Data Types	Heterogeneous (can mix types)	Homogeneous (all elements must be the same type)
Memory Layout	Pointers to scattered objects in memory	Contiguous block of memory
Performance	Slow; requires Python loops (for)	Fast; executed in optimized C code
Operations	Concatenation/Repetition (list + list)	Vectorized math (array + array)

In [ ]:
```
import numpy as np
```

In [ ]:
```
x = np.array([0, .5, 1., 2, 3, 4, 5, 6])
3 * x + 2
```

Array properties

- ndim: number of dimensions
- size: total number of elements
- shape: rows, columns, ...
- dtype: data type of elements
- nbytes: number of bytes occupied by data
- itemsize: number of bytes per element
    - nbytes = size * itemsize

In [ ]:
```
print("x.ndim  :", x.ndim)
print("x.size  :", x.size)
print("x.shape :", x.shape)
print("x.dtype :", x.dtype)
```

```
print("x.nbytes:", x.nbytes)
print("x.itemsize :", x.itemsize)
```

Let us look at some higher dimensional arrays (2D and 3D):

In [ ]:
```
# Create a 2D array (matrix)
data = np.array([[1, 2, 3],
                 [4, 5, 6]])

# Inspect fundamental array attributes
print(data.shape)  # Output: (2, 3) -> 2 rows, 3 columns
print(data.ndim)   # Output: 2      -> 2 dimensions
print(data.dtype)  # Output: int64  -> data type of elements
print(data.size)   # Output: 6      -> total number of elements
```

In [ ]:
```
# Create a 3D array (a stack of three 2x4 matrices)
data_3d = np.array([
    [[1, 2, 3, 4],
     [4, 6, 7, 8]],

    [[9, 10, 11, 12],
     [13, 14, 15, 16]],

    [[17, 18, 19, 20],
     [21, 22, 23, 24]]
])

# Inspect fundamental array attributes
print(data_3d.shape)  # Output: (3, 2, 4) -> 3 layers, 2 rows, 4 columns
print(data_3d.ndim)   # Output: 3         -> 3 dimensions
print(data_3d.dtype)  # Output: int64     -> data type of elements
print(data_3d.size)   # Output: 24        -> total elements (3 * 2 * 4)
```

How to read the shape of a multi-dimensional array:

- Axis 0 (size 3): The outermost dimension representing the depth or layers (3 distinct 2D matrices).
- Axis 1 (size 2): The middle dimension representing the rows within each matrix (2 rows).
- Axis 2 (size 4): The innermost dimension representing the columns in each row (4 columns).
- Quite a few NumPy functions ask you to specify which dimension it should be applied to

In [ ]:
```
data
```

In [ ]:
```
np.sum(data)  # Sum all elements if no axis is specified
```

In [ ]:
```
np.sum(data, axis=0)  # Sum along axis 0 (columns)
```

In [ ]:
```
np.sum(data, axis=1)  # Sum along axis 1 (rows)
```

Example use of 3D array: RGB images

In [ ]:
```python
import matplotlib.pyplot as plt
from PIL import Image  # Python Image Library

# Load image and ensure RGB mode
img = Image.open("cat_img.jpg").convert("RGB")
img.size
```

In [ ]:
```python
# Create an RGB image with random colors
img_array = np.array(img)
# Extract channels
red_channel = img_array[:, :, 0]
green_channel = img_array[:, :, 1]
blue_channel = img_array[:, :, 2]

# Create figure layout
fig, axes = plt.subplots(1, 4, figsize=(12, 4))

# Original RGB image
axes[0].imshow(img_array)
axes[0].set_title("RGB Image")
axes[0].axis('off')

# Red channel
axes[1].imshow(red_channel, cmap='Reds')
axes[1].set_title("Red Channel")
axes[1].axis('off')

# Green channel
axes[2].imshow(green_channel, cmap='Greens')
axes[2].set_title("Green Channel")
axes[2].axis('off')

# Blue channel
axes[3].imshow(blue_channel, cmap='Blues')
axes[3].set_title("Blue Channel")
axes[3].axis('off')

plt.tight_layout()
plt.show()
```

A note on numbers in NumPy

Property	**Python `int`**	**NumPy `int64`**	**Python `float`**	**NumPy `float64`**
Number Type	Integer	Integer	Real / Decimal	Real / Decimal

| Memory Size | Dynamic (≥28 bytes) | Fixed (8 bytes) | Fixed (≈24 bytes object) | Fixed (8 bytes raw) | Value Limits | Infinite (RAM bound)| -2^{63} to $2^{63}-1$ | $\approx \pm 1.79 \times 10^{308}$ | $\approx \pm 1.79 \times 10^{308}$ | | Overflow Behavior | Never overflows | Wraps around / Errors | Results in Inf | Results in inf | | Primary Purpose | Everyday Python scripts | Vectorized math & array speed | Everyday decimal math | High-performance linear algebra |

There are also other formats: `bool`, `int8`, `uint8`, `int16`, `uint16`, `int32`, `uint32`, `float8`, `float16`, `float32`, +++

Creating arrays

Numbers with fixed spacing: `linspace(a, b)`

- Creates array with 50 numbers from a to b
- Both a and b er included
- Fixed interval between numbers: $(b-a)/49$
- You get an array with 100 numbers using `linspace(a, b, num=100)`

In [ ]:
```
np.linspace(1, 50)
```

In [ ]:
```
np.linspace(0, 1, num=11)
```

In [ ]:
```
# Generate 100 evenly spaced points between -5 and 5
x = np.linspace(-5, 5, 100)

# Define the function y = x^2
y = x ** 2

# Plot the function
plt.figure(figsize=(8, 5))
plt.plot(x, y, label=r'$y = x^2$', color='blue', linewidth=2)

# Add labels, title, and grid
plt.xlabel('x', fontsize=12)
plt.ylabel('y', fontsize=12)
plt.title('Plot of y = x² using np.linspace', fontsize=14)
plt.grid(True, linestyle='--', alpha=0.7)
plt.legend()
plt.tight_layout()

# Show the plot
plt.show()
```

Numbers with logarithmic spacing: `logspace(a, b)`

- Creates array with 50 numbers from 10^a to 10^b
- Both 10^a and 10^b are included
- Logarithmic distance between numbers (fixed factor)

In [ ]:
```
np.logspace(-1, 1, num=10)
```

In [ ]:
```
# Generate values spaced evenly on a log scale
# Start exponent = 0 (10^0 = 1), stop exponent = 2 (10^2 = 100)
# 50 points between 1 and 100
x = np.logspace(0, 2, num=50, base=10.0)

# Simple function: y = x^2
y = x ** 2

# Plot
plt.figure(figsize=(8, 5))
```

```
plt.plot(x, y, marker='o', linestyle='-', color='blue', label=r'$y = x^2$')

# Set log scale for x-axis to match spacing
plt.xscale('log')

# Labels and title
plt.xlabel('x (log scale)')
plt.ylabel('y')
plt.title('Plot of y = x² using np.logspace')
plt.grid(True, which="both", ls="--", linewidth=0.5)
plt.legend()

# Show plot
plt.show()
```

Random numbers

- Use random number generator to fill the array
- Generates an array filled with random floating-point values sampled from a uniform distribution over the interval $[0.0, 1.0)$.

In [ ]:
```
np.random.rand(2, 2)
```

Some other functions generating arrays

- These functions take shape as first argument, data type is optional (default is `float64`)
- `zeros()`: array of zeros of given size
- `ones()`: array of ones of given size
- `empty()`: array with uninitialized data
- `arange()`: similar to range, really useful only for arrays of integers
- *_like(): create array with same shape as argument
  - `zeros_like(a)`: creates an array of zeros with the same shape as a

In [ ]:
```
np.ones([2, 3])  # 2x3 matrix of 1.0
```

- Note that empty allocates a block of memory for the specified shape and data type, but does not write any values to it.
- The entries you see are whatever random bits happened to exist in that memory block at the moment of allocation.

In [ ]:
```
np.empty(3)  # 3x1 vector filled with some numbers
```

Working with data in multiple dimensions

- We will stick to one and two dimensions here
- Key concepts:
  - Operations are applied element-wise if both operands have same size
  - Otherwise, NumPy attempts to *broadcast*

- The smaller operand is replicated as many times as necessary to create an array that matches the larger operand
- The operation is then applied element-wise
- This allows NumPy to perform operations between arrays of different shapes without taking up unnecessary memory

Example: array and array of same size

In [ ]:
```
a = np.array([[1, 2], [3, 4], [5, 6]])
b = np.array([[10, 20], [30, 40], [50, 60]])
print(a)
print("Shape of a: ", a.shape)
print(b)
print("Shape of b: ", b.shape)
```

In [ ]:
```
a + b
```

In [ ]:
```
# A reminder about how addition of Python lists work
list_1 = [1, 1, 1, 1]
list_2 = [1, 2, 3, 4]
list_1 + list_2
```

In [ ]:
```
# Addition of NumPy arrays is element-wise
array_1 = np.array([1, 1, 1, 1])
array_2 = np.array([1, 2, 3, 4])
array_1 + array_2
```

In [ ]:
```
a * b
```

- Note: This is element-wise multiplication, not matrix multiplication in mathematical sense!

Mathematical matrix-matrix/matrix-vector multiplication

- First factor must have as many columns as the second has rows
- a and b are both 3x2, so we get an error if we try matrix multiplication here

In [ ]:
```
a @ b  # mathematical matrix multiplication
```

- If we transpose b (flip about diagonal) to make it 2x3, the dimensions are compatible

In [ ]:
```
b.T  # b transposed
```

In [ ]:
```
a @ b.T
```

Example: array and scalar

In [ ]:
```
a
```

In [ ]:
```
a + 100
```

In [ ]:
```
a * 2
```

In [ ]:
```
a ** 2
```

Example: 2D- and 1D arrays (matrix and vector)

In [ ]:
```
c = np.array([100, 200])  # 1D array (vector)
c
```

In [ ]:
```
a + c
```

- c has been added to each row of a: broadcasting
- What happens for 3-element array?

In [ ]:
```
d = np.array([10, 20, 30])
d
```

In [ ]:
```
a + d
```

- NumPy sees d as 1x3 vector, so shape does not fit
- Need to transpose first!
- But because d has only one dimension, transposing it has no effect

In [ ]:
```
d.T
```

- We need to define a two-dimensional array that has one row and three columns (note the extra brackets!) that can be transposed

In [ ]:
```
dd = np.array([[10, 20, 30]])
dd
```

In [ ]:
```
dd.T
```

In [ ]:
```
a + dd.T
```

Arrays vs Lists

- NumPy arrays differ in several ways from Python lists, especially
  - arrays have fixed size
  - all entries in an array have the same data type (exceptions apply)
- This makes arrays much more efficient in time and space than lists
- How would you double all elements in a Python list? And how would you do this for a 1D NumPy array?

In [ ]:
```
# Python approach:
py_list = [1, 2, 3, 4]
py_doubled = #
print(py_doubled)

# NumPy approach (vectorized):
np_array = np.array([1, 2, 3, 4])
np_doubled = #
print(np_doubled)
```

Performance comparison

- List and array with 1.000.000 entries
- Compute sine of all elements

In [ ]:
```
lx = list(range(0, 1000000))
ax = np.array(lx)
```

In [ ]:
```
# Calculate the sine of every element in the list lx
import math  # to use the math.sin() function
%timeit sinlx = [math.sin(x) for x in lx]
```

In [ ]:
```
# Calculate the sine of every element in the array nx
%timeit sinax = np.sin(ax)
```

- We use `%timeit` to compare computation times here (benchmarking)
  - This is an example of Jupyter/IPython magic
  - It runs everything that follows on the line, taking time with a stopwatch
  - repeats the timing several times to get statistics on fluctuations
  - to ensure reasonably accurate timing, it executes (loops) the command many times until runtime is at least 0.2s

In [ ]:
```
%timeit?
```

In [ ]:
```
sinlx = [math.sin(x) for x in lx]
sinax = np.sin(ax)
all(sinlx == sinax)
```

- NumPy is significantly faster than list comprehension
- This applies mainly when we apply NumPy functions to large arrays
- For scalars, `math` functions are usually faster than `numpy` functions

Arrays slices are view: An important difference from lists

- Array slices are so-called views:
  - If a is a Numpy array, and `b = a[m:n:s]`, then b is a *view* of array a
  - a and b are distinct objects (i.e., `id(a) != id(b)`)
  - a and b are both of type `numpy.ndarray`

    - a and b share the same underlying data: b just restricts what we see of the data in a
    - Therefore, any change to data in b also affects data in a

In [ ]:
```
# Lists
ls1 = [1, 2, 3, 4, 5]
ls2 = ls1[:3]
ls2[0] = 10
print("lists:")
print(ls1)
print(ls2)

a = np.array([1, 2, 3, 4, 5])
b = a[:3]
print(id(a))
print(id(b))
b[0] = 10
print("arrays:")
print(a)
print(b)
```

- For a normal Python list `l`, `l[:]` is a copy of the elements in `l`, i.e., a new list object, that can be manipulated without affecting `l`.
- For an array `a`, `a[:]` is a view of a with the same underlying data as `a` and changing data through this view will change the data in `a`.
- The issues with copying that we discussed earlier for lists are also precent for arrays

In [ ]:
```
b = a.copy()  # Creates a copy rather than a view
b[0][0] = 10
print(a)
print(b)
```

Pandas series and dataframes

Pandas

- Powerful and widely popular package for data processing
- Originally geared towards applications in finance
- Focus on managing data
- Can be considered Python's answer to spreadsheets
- Similar to R's data frames
- Some support for data analysis and visualization

Key concepts

- Series, roughly comparable to single columns in a spreadsheet
- DataFrame, roughly comparable to an entire table or spreadsheet
- Builds on NumPy, but
    - adds names to data
    - provides powerful high-level tools to work with tables

NumPy or Pandas?

- NumPy for "pure" mathematics
- Pandas when you need metadata such as column names and indexes
- Both NumPy and Pandas has a lot (!) of build in functions, explore them!
  - Mathematics
  - Statistics
  - Visualisation

Example: Moving averages in NumPy and Pandas

In [ ]:
```python
def numpy_moving_average(data, window_size):
    # Create a normalized weight array: e.g., for window 3 -> [1/3, 1/3, 1/3]
    weights = np.ones(window_size) / window_size
    # Convolve with mode='valid' to calculate averages where full windows exist
    return np.convolve(data, weights, mode='valid')


data = np.array([10, 20, 30, 40, 50, 60, 70])
result = numpy_moving_average(data, window_size=3)

print(result)
```

In [ ]:
```python
import pandas as pd


def pandas_moving_average(data, window_size):
    series = pd.Series(data)
    # Calculates the rolling mean across the specified window length
    return series.rolling(window=window_size).mean()


data = [10, 20, 30, 40, 50, 60, 70]
result = pandas_moving_average(data, window_size=3)

print(result)
```

- You can think of convolution as a mathematical way of applying a sliding filter or template across a stream of data
- The NumPy version is very fast, and is good for situations where speed and efficient memory use is a priority
- The Pandas version is especially good for time series, where you have a Series or DataFrame with time as index